

ES 333 Microprocessors and Embedded Systems lab

Project

IoT-based Wireless Charging Station for Electric Vehicles (EVs)

Problem Statement:

The rising use of electric vehicles (EVs) comes with challenges involving charging infrastructure and efficiency. Plug-in charging technology involves physical contacts, which causes inconvenience and loss of mobility. To overcome such challenges, a wireless power transmission system can promote user convenience, increase energy efficiency, and decrease reliance on stationary charging points.

Theory:

Inductive coupling and magnetic resonance allow power transfer between two coils without physical connection. Using resonant frequency ($f = 1/2\pi\sqrt{LC}$), energy can be transferred efficiently over short distances.

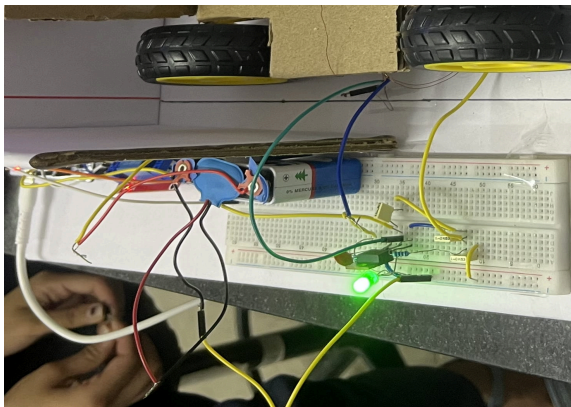
Key Components Used:

- STM32 Board
- LCD Display (16x2)
- Relay Module
- MOSFETs
- Power Transmission Coil
- Receiver Coil (Mounted on EV)
- 7805 Voltage Regulator (2)
- IR Sensor

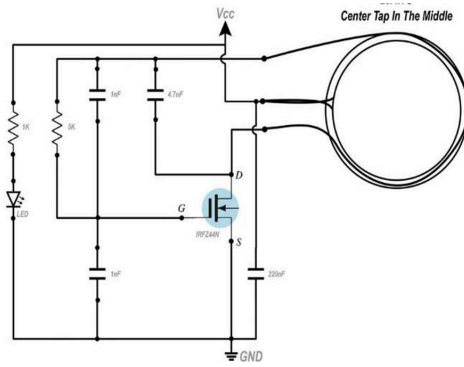
Hardware Implementation:

Split into two parts:

1. **Transmitter Circuit** (Components: MOSFET, Primary Coil, 9V Supply)

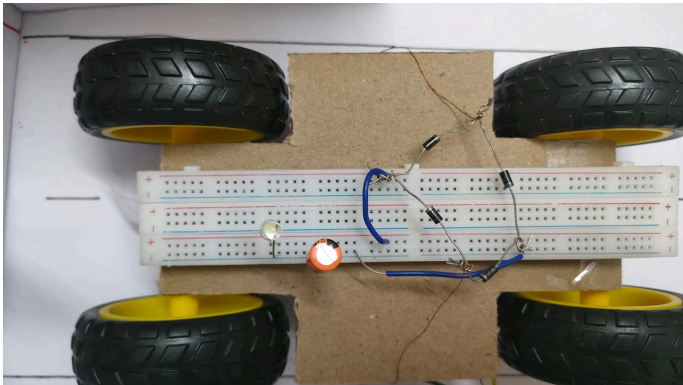


Hardware Model of Transmitter Circuit

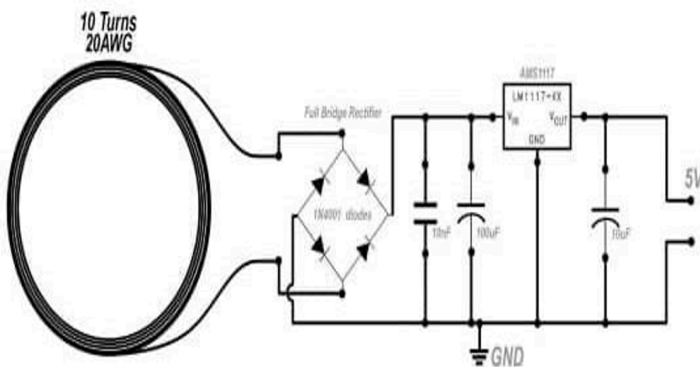


Wireless Mobile Charger Diagram (Transmitter)

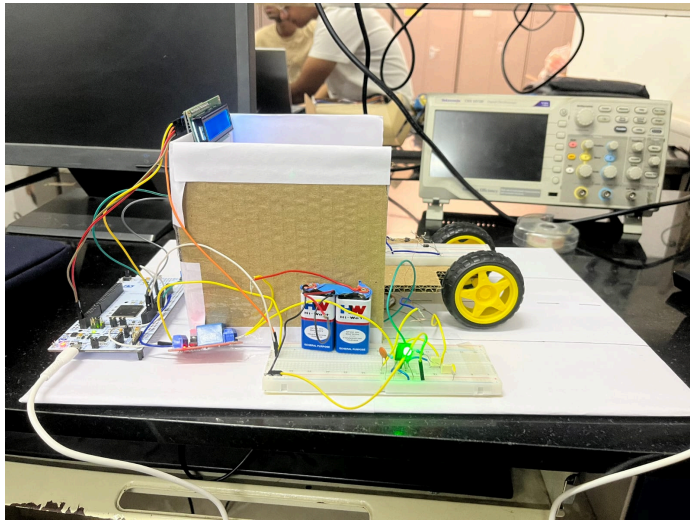
2. Receiver Circuit (Bridge rectifier, Capacitors, Voltage Regulator 7805)



Hardware Model of Receiver Circuit



Wireless Mobile Charger Diagram (Receiver)



Overall Hardware Model

Working Principle:

The proposed system operates on the principle of inductive power transfer (IPT), allowing electrical energy to be transmitted wirelessly between two coils one connected to a power source (transmitter) and the other to the load (receiver). When a vehicle approaches the charging area, an IR sensor detects its presence and sends a signal to the STM32 microcontroller. Upon receiving this signal, the STM32 activates a relay module, which in turn connects the transmitter circuit to the DC power supply.

The STM32 also generates a high-frequency that drives a MOSFET, causing current to flow through the primary coil. This current produces an oscillating magnetic field, which extends a few centimeters around the coil. When the secondary coil, mounted beneath the vehicle, comes within this field, an alternating voltage is induced in it due to electromagnetic coupling.

This induced AC voltage is then passed through a bridge rectifier to convert it into DC. To ensure stable output, the signal is filtered using capacitors and then regulated to 5V DC using a 7805 voltage regulator. The final output can be used to power an LED, a small motor, or charge a battery.

Additionally, the STM32 microcontroller can monitor the system's status and transmit real-time data such as voltage, current, or charging status. A 16x2 LCD display is also used to show local charging status such as "Charging", "Idle", or "No Vehicle Detected". This intelligent combination of sensing, control, wireless energy transfer, and IoT integration forms a smart and efficient solution for wireless EV charging.

Results / Output:

The system worked properly in many areas. The IR sensor was correctly detecting the presence or absence of a vehicle. When the sensor detected a vehicle, it sent a signal to the STM32, which then activated the relay. The relay turned on when the vehicle was present and turned off when it left. This ensured that power was supplied only when needed. The MOSFET-based circuit converted DC to AC correctly and also the full bridge rectifier on the receiving side converted the AC back to DC, allowing the DC power used by vehicle.

But, the main issue was in the wireless power transfer between the coils. Even though the other parts worked as expected, the power transfer was very weak. This was likely because the distance between the coils was very large, which reduced the strength of the magnetic field and made the transfer inefficient. Also, the coils were made using very thin copper wire, which caused extra resistance and power loss. These were the two reasons for the poor energy transfer, and as a result, the wireless charging did not work properly.

Applications:

- Real-world EV roads with embedded charging pads
- Smart Parking Stations with automatic wireless charging
- Public charging infrastructure integrated with IoT and billing

References:

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[3]S. R. Khutwad and S. Gaur, "Wireless charging system for electric vehicle," *2016 International Conference on Signal Processing, Communication, Power and Embedded System (SCOPES)*, Oct. 2016, doi: <https://doi.org/10.1109/scopes.2016.7955869>.

[4]M. Amjad, M. Farooq-i-Azam, Q. Ni, M. Dong, and E. A. Ansari, "Wireless charging systems for electric vehicles," *Renewable and Sustainable Energy Reviews*, vol. 167, p. 112730, Oct. 2022, doi: <https://doi.org/10.1016/j.rser.2022.112730>.

[5]S. Inamdar and J. Fernandes, "Review of Wireless Charging Technology For Electric Vehicle," Nov. 2022, doi: <https://doi.org/10.1109/piicon56320.2022.10045150>.